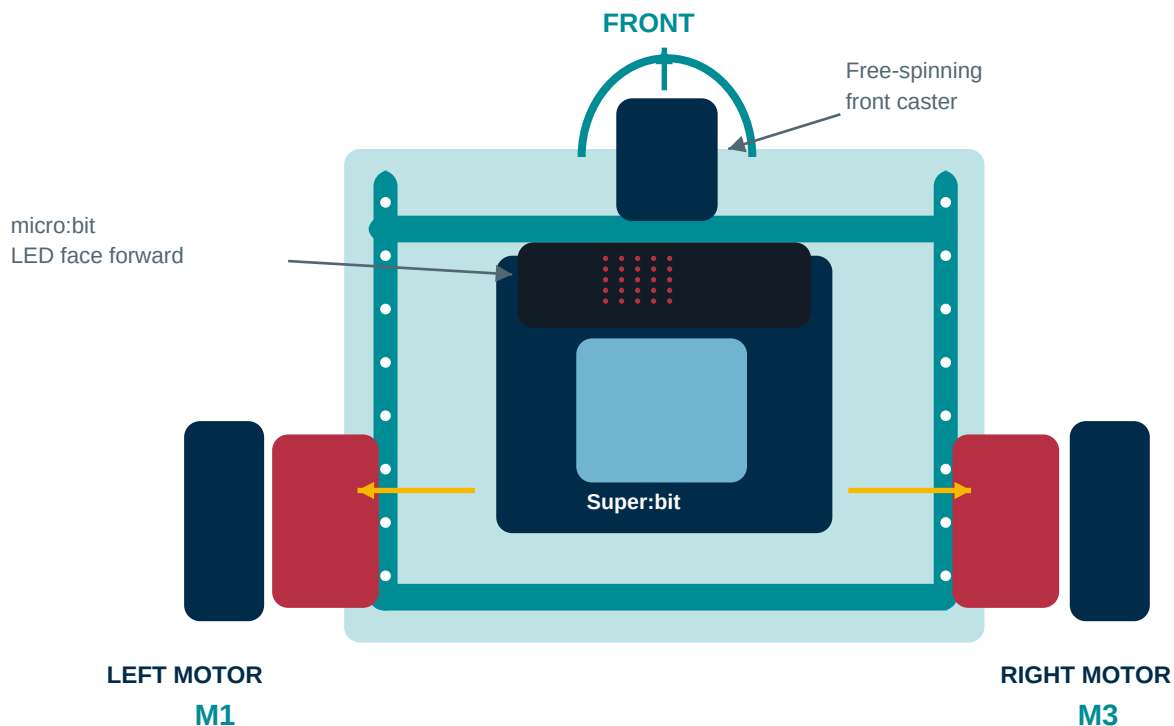




Build a voice-controlled rover

A complete, illustrated build-and-drive guide for the three-wheel ThinkerLab rover.

Two rear drive wheels. One free-spinning front support. One iPhone.



Layout illustration - not a parts-scale drawing. View from above; front is toward the LED face.

Your route from box to moving robot

BUILD	INSTALL	DRIVE
Parts + chassis Pages 2-4	Firmware + iPhone app Pages 5-6	Connect + your control mode Pages 7-10

Before you start

Use a Mac with Xcode and a micro:bit V2. This is a source-installed iPhone app, not an App Store download. Start with the wheels lifted.



01 / Gather your parts

A stable rover starts with the correct motors, connectors and a supported phone.

Quantity	What you need
1	Yahboom Building:bit Super Kit / Super:bit board, plus a micro:bit V2. Check which pieces your kit edition includes.
2 + 2	Two red DC drive motors and two matching wheels. Use equal wheel diameters.
1	A free-spinning front wheel in a swivel fork (caster), or a ball caster with a compatible bracket. Add one if your kit lacks it.
As needed	Rigid beams/base plate, cross-braces, axles/bushes and compatible pins or fasteners. No soldering is required for the standard connectors.
1 set	The correct Super:bit battery and its approved charging arrangement; data cables for micro:bit and iPhone; a stable phone stand.
1 each	Mac, Apple Account and iPhone. Internet is needed for software downloads/signing.

Software and compatibility

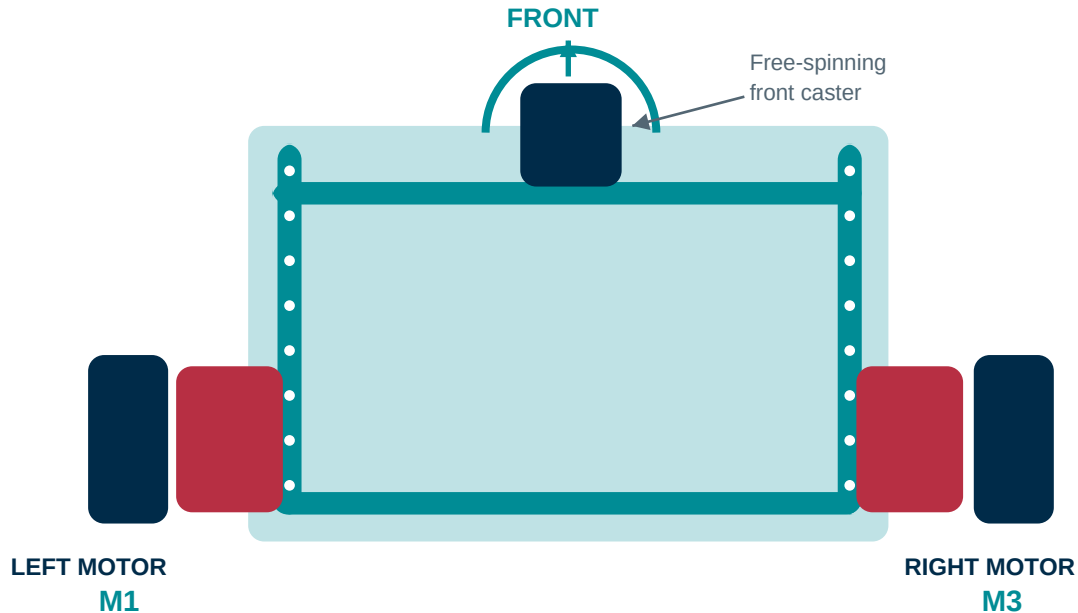
These guides target app **1.7.2 (build 15)**, built with Xcode 26.5 and tested on iPhone 17 Pro / iOS 26.6. The project targets iOS 17+, but older devices have not been validated. Use Xcode 26 or newer with an SDK/device-support version compatible with your phone.

Voice needs microphone and on-device English (Australia) speech recognition availability. Flexible AI parsing additionally needs iOS 26+ and an available Apple Intelligence model; basic commands do not.



02 / Build a rigid three-wheel chassis

A custom triwheel layout using compatible kit parts; exact brick lengths can vary.



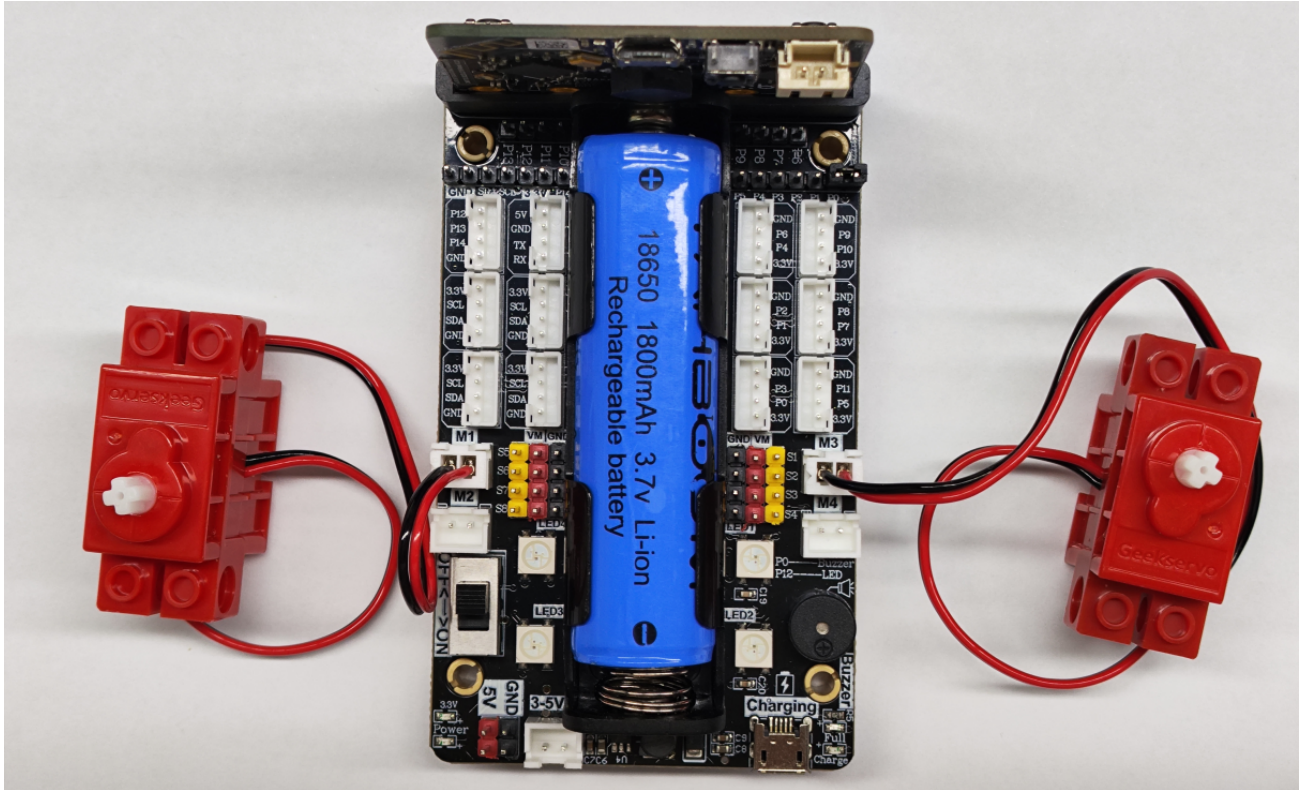
- 1 Make a braced base**
Join two parallel beams with front and rear cross-braces. Add a plate or a second brace so the frame cannot twist when you gently flex opposite corners.
- 2 Fit the rear motors and wheels**
Mount one DC motor on each rear side. Keep shafts level, parallel and facing outward. Fit the two drive wheels; leave a small gap so tyres and hubs do not rub on beams.
- 3 Add the front support**
Centre the caster at the front. Fit its fork/axle loosely enough to roll freely but securely enough not to fall out. A swivel or ball caster makes rotation easier; a fixed front wheel may scrub sideways.

CHECK: all three wheels touch a flat surface; the chassis sits level; rear wheels rotate freely; the caster swivels through a full turn. Adjust bracket height before adding electronics.



03 / Mount, connect and power

Turn the board OFF and disconnect charging/USB power before moving connectors.



Yahboom reference photo: left motor on M1, right motor on M3. Match the printed labels on your board; this is not a photo of the custom chassis. [2]

4

Secure the board and insert micro:bit

Use the Super:bit mounting holes and compatible pins/standoffs. Keep metal away from exposed contacts. Insert the micro:bit fully in the edge connector, oriented as the manufacturer shows; keep its LED face toward your chosen front.

5

Connect the drive motors

Viewed from behind, looking forward: plug the LEFT motor into M1 and RIGHT motor into M3. Use the keyed two-wire plugs without forcing them. Leave M2, M4 and the servo sockets unused. Route leads clear of the wheels.

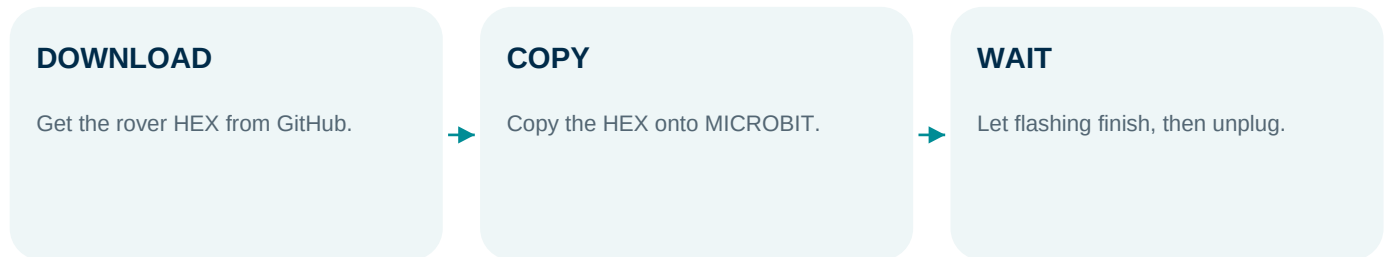
Battery and charging

Use only the battery type/polarity specified for your board revision. The pictured board uses one 3.7 V 18650 cell. Charge through the Super:bit charging port, not the micro:bit USB port; do not drive while charging. Keep its switch OFF until testing. [3]



04 / Put rover firmware on micro:bit

Firmware is the small program that receives Bluetooth commands and controls the motors.



- 1 Download the correct program**

Open [the firmware download](#) and save **microbit-rover.hex**. Or open [GitHub / downloads](#) and use the file download button. This replaces the program already on micro:bit. Do not use the vendor demo firmware or the LED-only example.
- 2 Connect micro:bit to the Mac**

Leave motor power off. Connect a USB data cable to the micro:bit itself. Finder should show a drive named **MICROBIT**. A charging-only cable will not work.
- 3 Copy and wait**

Drag microbit-rover.hex onto MICROBIT. Wait for the transfer and the flashing activity LED to finish. If a FAIL.TXT file appears, read it and retry with a known data cable. The supplied firmware shows a small diamond while disconnected.
- 4 Disconnect the programming cable**

The rover will run wirelessly once its board battery is switched on. A USB cable in the micro:bit may power only the controller; it is not a substitute for the board motor supply.

Optional: build the HEX from source

Install Node.js/npm, download/extract the repository, and run these commands in its folder. The build script uses the MakeCode cloud compiler, so internet is required.

```
bash scripts/build-firmware.sh
```

```
cp -X .build/microbit-rover.hex /Volumes/MICROBIT/
```

The ready-made HEX is supplied for this guide; no Arduino or ESP32 firmware is needed.



05 / Install Microbit Link on iPhone

Use your own Apple Account and signing identity. There is no App Store or TestFlight link.

- 1 Get the current source**

Open [the project on GitHub](#). Choose **Code > Download ZIP**, then extract it on your Mac. Open **MicrobitLink.xcodeproj** from the extracted folder. Use version 1.7.2 or later; the older release still has Enable buttons and no voice screen.
- 2 Set up Xcode signing**

Install a compatible Xcode from Apple and open it once to finish setup. In Xcode Settings > Apple Accounts, sign in. Select the blue project, then the MicrobitLink target > Signing & Capabilities. Enable automatic signing and choose your own Team.
- 3 Choose an app identifier**

Replace org.example.microbitlink with a unique identifier such as **com.yourname.rover**. Keep that identifier for later updates; changing it creates a separate app with separate preferences.
- 4 Trust and prepare the phone**

Connect and unlock iPhone, accept Trust prompts and select it as Xcode's run destination. If requested, enable Settings > Privacy & Security > Developer Mode, restart and confirm. Developer Mode may appear only after pairing with Xcode. [4,5]
- 5 Build, install and open**

Click the Run triangle (Command-R). Wait for a successful build and installation. Follow any developer-trust prompt on iPhone; then open **Microbit Link** and allow Bluetooth. The app screen is titled **Microbit Rover**. [4]

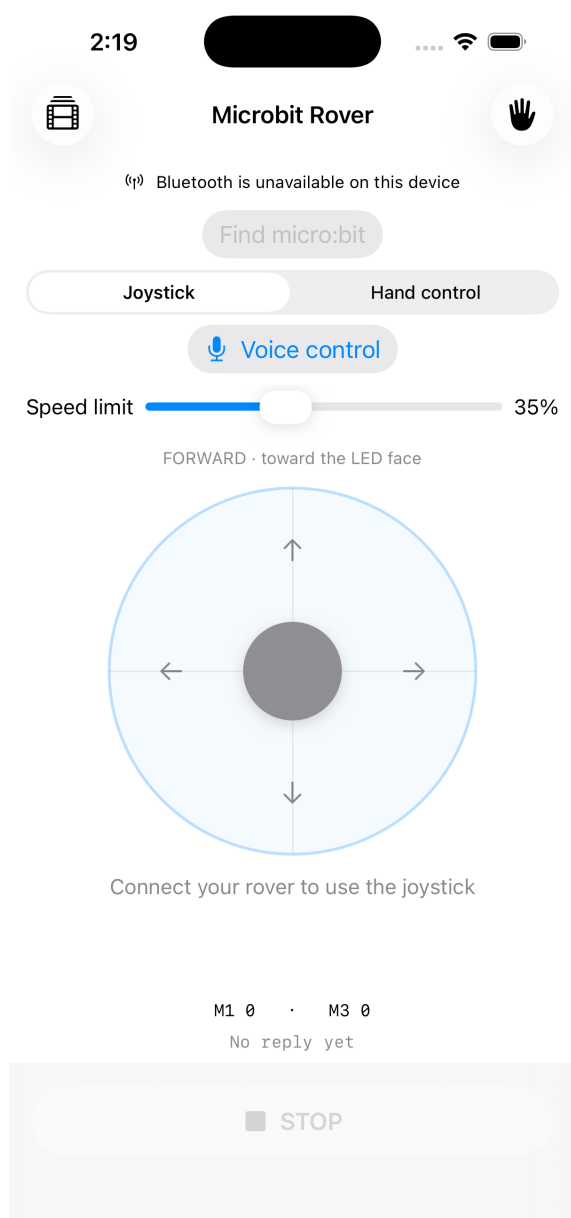
Keep the installation working

A free Personal Team installation expires after seven days; rebuild and reinstall with the same identifier to renew it. If Xcode cannot support your iOS version, update Xcode/macOS as required before retrying. An iPhone alone cannot perform this source-install workflow. [6]



06 / Connect and check wheel direction

Do this once with joystick control before trying gestures or speech.



Actual app UI in the iPhone simulator. Bluetooth is unavailable there; your physical phone shows nearby boards after scanning.

1

Lift the drive wheels

Support the chassis securely so both rear wheels are off the table. Switch on the Super:bit battery supply.

2

Connect inside the app

Tap Find micro:bit, then select your board. Wait for Connected - controls ready. Connect in this app, not the iPhone Bluetooth Settings list.

3

Try a tiny forward touch

Select Joystick and use the default 35% limit. Touch briefly above centre, then release. Both wheels should drive toward the marked front.

4

Correct the mapping

Tap STOP > Wheel setup. Keep "M1 is the right wheel" OFF for M1-left/M3-right. Reverse only a motor that turns backward. Tap Done and test again.

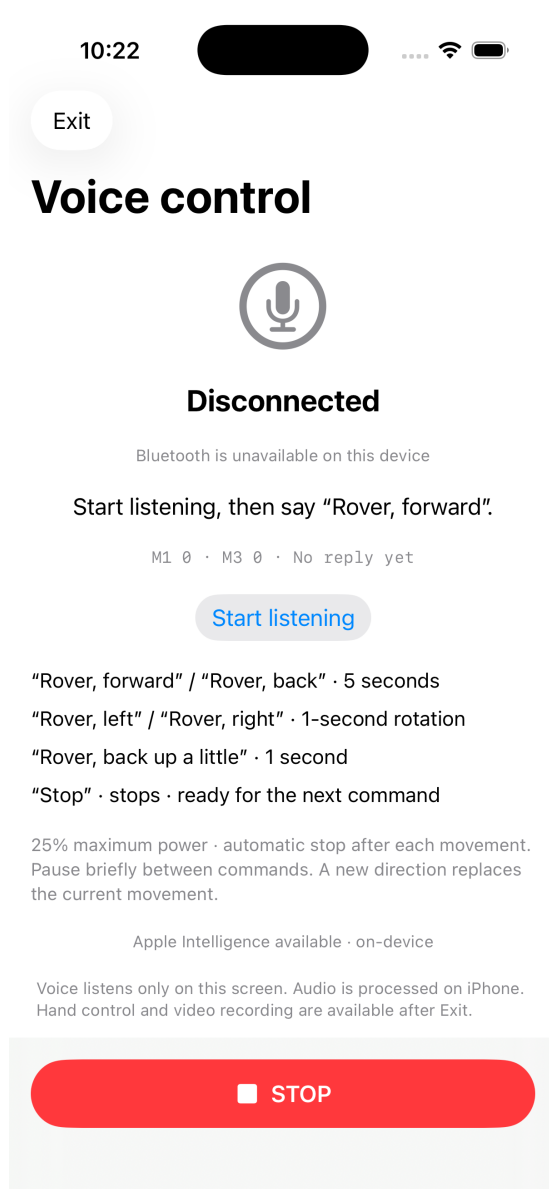
Pass this check before floor driving

Release stops both wheels. Left/right rotate the expected way. STOP and either micro:bit button stop movement. A new touch is needed after Stop. There is no separate Enable step: the app handles ARM when fresh input arrives.



07 / Drive with your voice

Start on a clear floor at low speed; keep STOP and the board power switch accessible.



Actual 1.7-series simulator UI. The phone reports its own speech/model availability.

1

Open Voice control

Connect the rover first, then tap Voice control on the main screen.

2

Start listening

Tap Start listening and allow Microphone and Speech Recognition. Keep the phone unlocked on this screen.

3

Say one complete command

Say "Rover, forward" and pause briefly. Read Heard and the accepted action. Movement ends automatically after five seconds.

4

Turn or stop

Say "Rover, right" for a one-second rotation. Say "Stop" to cancel immediately when recognised. No Enable button is needed for the next fresh command.

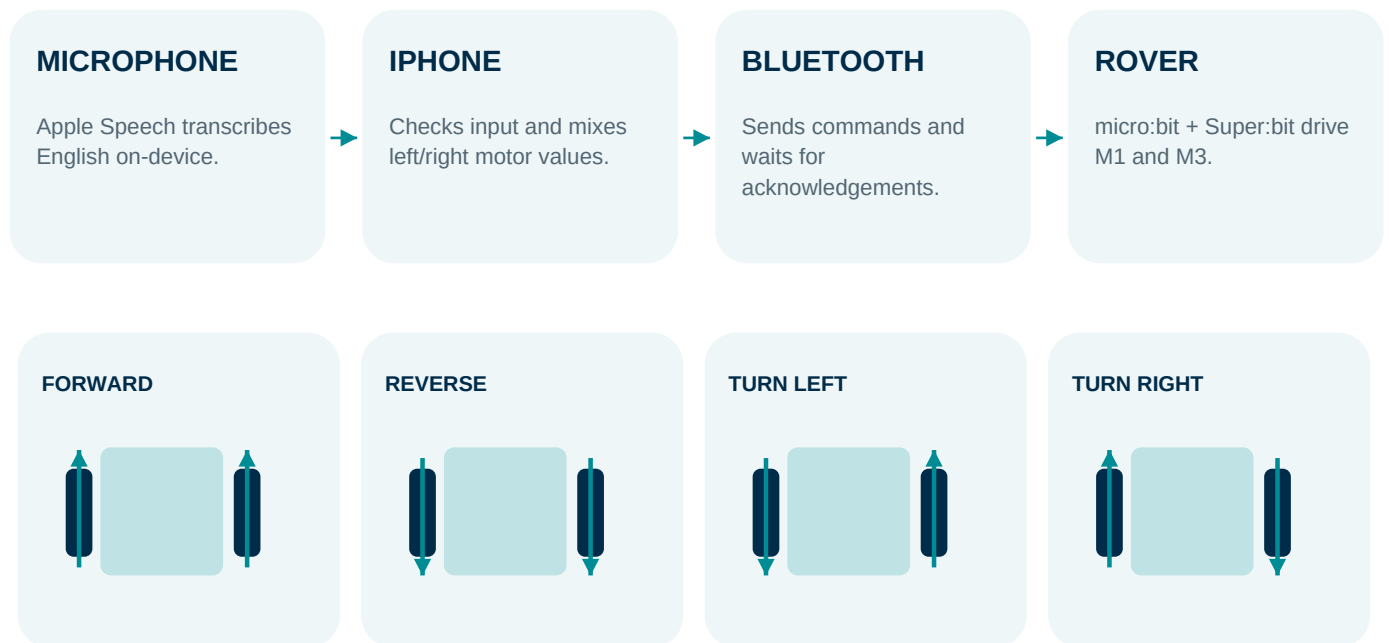
Try this sequence

"Rover, forward" > wait for it to stop > "Rover, left" > "Rover, right" > "Stop" > "Rover, back". Begin with wheels lifted. A new direction replaces the current action; phrases are not queued.



08 / Understand the technology

The iPhone interprets your input. The micro:bit runs the motor program.



Speech recognition and intent are separate

Apple Speech converts microphone audio into words locally. A strict parser handles directions and polite can/could/would/will-you forms. Stop words are checked in partial transcripts without waiting for AI. [8]

Apple Intelligence is optional

If available on iOS 26+, the local Foundation Models model can classify flexible wording. The app still requires one explicit direction and the Rover prefix; AI cannot invent power, duration or a route. [9]

Bounded actions, no old command queue

Forward/reverse last five seconds; turns one second. "A little" shortens movement. The app consumes each transcript segment once. New speech or Stop invalidates pending AI; results older than three seconds are discarded.

The stop chain

The app sends motor updates at 10 Hz, with one outstanding reply. A reply timeout around 350 ms triggers Stop/disconnect. Firmware separately disarms after 400 ms without a valid drive update. These are software checks; wheels can coast, and they do not detect obstacles.



09 / Your voice command reference

Use short, repeatable trials so you can tell what changed.

Say this	Expected action
Rover, forward / go forward	Forward for 5 seconds
Rover, back / reverse / back up	Reverse for 5 seconds
Rover, left / turn left	Rotate left for 1 second
Rover, right / turn right	Rotate right for 1 second
Rover, can you turn right a little	Rotate right for 0.5 seconds
Rover, turn left a little	Rotate left for 0.5 seconds
Rover, back up a little	Reverse for 1 second
Stop / halt / freeze / cancel	Cancel movement; no Rover prefix needed

What the screen tells you

- **Heard:** what speech recognition transcribed, not proof that a command was accepted.
- **Action + seconds:** the interpreted direction and bounded duration. M1/M3 show requested outputs; the reply shows what the rover reported.
- **Interpreting on iPhone:** optional AI is working. Use a simpler command if it times out.
- One direction per phrase. Requests for angles, distances, custom times, conditions or multiple actions are rejected. "Turn 90 degrees" is not supported.

Voice output is capped at 25% (or a lower saved limit). A half-second turn may be small on a grippy surface. First compare "Rover, right" with "Rover, right a little"; do not assume a recognised phrase proves the wheels turned.



10 / Troubleshoot in the right order

Stop first. Check the visible evidence before changing wiring or settings.

Symptom	Check / next action
No MICROBIT drive	Try a known USB data cable and another port. Connect to micro:bit, not the board charging port. Check FAIL.TXT after a failed flash.
Xcode will not install	Unlock/trust the phone; select the correct device and your own Team/unique bundle ID. Update Xcode if the phone OS is unsupported. Renew an expired Personal Team install.
No rover found	Switch on the board; allow Bluetooth in iPhone Settings; scan in Microbit Link. Disconnect other clients. Reflash the rover HEX if vendor firmware is still installed.
Connected but motors do not run	Check the Super:bit switch, charged correct battery, M1/M3 connectors and wheel clearance. USB in micro:bit alone may not supply the motors. Test joystick first.
Directions are wrong	STOP > Wheel setup. Confirm M1-left/M3-right. Reverse only the motor that is backward. Swap wheels only if physical left/right really are exchanged.
Heard changes, but no motion	Read the exact Heard text and action/status. Start with "Rover, right". Version 1.7.2 directly accepts "Rover, can you turn right a little". A phrase without Rover is ignored except Stop words.
No speech / AI unavailable	Check microphone and speech permissions. On-device English recognition must be available. Apple Intelligence is optional for basic commands. Pause/start listening after an interruption.
A little turn is barely visible	Compare the one-second turn with the half-second turn, wheels lifted first. Check requested M1/M3 values and tyre/caster friction. Output percentage is not guaranteed torque.

If the problem remains: record the app version, exact status/reply, control mode, motor values, and whether the wheels moved while lifted. Share a wiring photo showing M1/M3 labels and describe one repeatable test.



Keep this page beside the rover

Session checklist, recovery actions and source links.

Every session

1. Check wheels, loose leads and battery. 2. Connect in Microbit Link. 3. Test a short movement and Stop with wheels lifted. 4. Put the rover on a clear floor. 5. Keep the app foreground and watch the rover. 6. Finish with STOP, switch off board power, then store safely.

Know how to stop

Joystick: release. Gesture: open pinch or remove hand. Voice: say Stop, or use the screen if speech is delayed. Red STOP and either micro:bit button cancel movement. If software is unresponsive, switch off board motor power. Nothing here senses a cliff or obstacle.

Downloads and references



[1] [Source, app and firmware](#)

[2] [Yahboom kit and motor wiring](#)

[Yahboom M1/M3 photo source](#)

[3] [Board charging instructions](#)

[4] [Apple: run on a physical device](#)

[5] [Apple: Developer Mode](#)

[6] [Apple: Personal Team and renewal](#)

[8] [Apple: speech recognition](#)

[9] [Apple: Foundation Models](#)

[ThinkerLab](#)

GitHub: github.com/ibut-bot/iphone-microbit-rover

Kit: yahboom.net/study/buildingbit-super-kit

Brand: thinkerlab.com.au

Edition notes: app 1.7.2 / build 15, source snapshot a71c3e5. Joystick and gesture use were confirmed on the existing rover. Voice forward and ordinary right-turn were confirmed; the latest polite-phrase fix passed automated tests and still needs user confirmation. Triwheel drawings are a proposed custom layout, not a verified kit bill of materials.

Image credits: ThinkerLab logo from its website; motor/board photograph from Yahboom. App screenshots are simulator captures. All labelled chassis, control and data-flow diagrams were created for these guides. Refer to the linked manufacturer instructions for your exact board revision.